

Measuring Validator and Sandboxed-Execution Coverage for LLM-Generated NetOps Artifacts: A Paired Confirmatory Study

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Abstract—We preregistered and executed a paired confirmatory experiment (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e) that measured the marginal detection benefit of adding sandboxed emulation to a deterministic validator for LLM-generated intent→configuration artifacts. Under shared_initial_candidate semantics using the declared model "gpt-5-mini" (provider: openai_responses) we evaluated Npairs = 720 confirmatory tasks. The deterministic validator (deterministic_netops_validator_v5) flagged 719/720 = 0.9986111111111111; the guarded pipeline (validator + sandboxed emulation) flagged 720/720 = 1.0. Paired difference (guarded - baseline) = 0.001388888888888884; exact McNemar two-sided p = 1.0; 95% bootstrap percentile CI (10000 resamples, seed = 1729) = [0.0, 0.004166666666666667]. Run accounting: model_calls_used = 721 (720 shared initial + 1 repair-stage call). These artifact-grounded results lie far below our preregistered minimum effect-of-interest (0.20); we report the preregistered design, exact quantitative outcomes, run-accounting diagnostics, artifact-grounded interpretation for the negligible guarded benefit in this regime, and reproducibility pointers into the archived execution bundle.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent into device configuration and NetOps workflows. Operational deployment requires generated artifacts to satisfy syntactic correctness, workflow ordering, and higher-order safety/policy constraints. A common mitigation architecture is generate→validate→(repair)→execute, sometimes augmented by sandboxed emulation to reveal runtime-only failures that static validators miss. Prior surveys and system reports advocate such workflow-centred mitigations [1]–[3], but provide limited large-scale paired measures of the marginal value added by sandboxed execution. Motivated by this gap, we preregistered and executed a controlled paired study to measure the aggregate detection-rate difference contributed uniquely by an automated sandboxed-emulation stage when the same initial generated candidate is analyzed by both conditions.

Research question and contributions. Our preregistered research question was: for synthetic intent→configuration tasks under shared_initial_candidate semantics, what is the paired detection-rate difference (guarded - baseline) between (A) a deterministic validator-only pipeline and (B) the same deterministic validator followed by automated sandboxed emulation? Contributions: (1) a machine-readable preregistration and faithful autonomous execution; (2) an artifact-grounded

paired analysis on Npairs = 720 with complete accounting; (3) an artifact-grounded interpretation explaining a near-zero marginal effect; (4) concise reproducibility pointers into the archived bundle and prescriptive boundary conditions where guarded stages are likely to matter.

II. RELATED WORK

Workflow-centred mitigations and verification tools. Recent survey and architectural reports document the practical trend of combining LLM generation with deterministic validators, emulators, and repair loops to raise operational assurance for NetOps artifacts [1]. These works emphasize that correctness in intent→configuration pipelines is multidimensional: syntactic parsing, command-ordering/workflow constraints, invariants (e.g., keep-one-uplink-up), and dynamic runtime properties may each require different V&V primitives. Tool-focused case studies highlight the evolution and practical limits of static verifiers: mature configuration-analysis systems have progressively expanded rule-sets and topology-aware checks but still exhibit gaps for runtime-dependent behaviors and stateful sequences [2]. That evolution shows why researchers and operators often layer an emulator or sandbox after static validation: emulation can exercise temporal behaviors, device state transitions, and interactions that are difficult to capture exhaustively as static rules.

Empirical and prototype evidence. Prototype systems that integrate generation with verification and execution-harnesses report heterogeneous gains: in some task regimes a verifier alone catches most dangerous errors, while in others emulation uncovers subtle runtime failures or environment-specific side effects that static rules miss [1], [3]. The practical implication drawn across this literature is that the marginal utility of a guarded (validator+emulation) pipeline depends on (a) validator coverage relative to the task distribution, (b) the fidelity of the emulator, and (c) the distributional properties of the input intents (e.g., constrained, parser-validated intents versus noisy natural-language requests). These prior empirical and system-level observations motivated a paired experimental design: by reusing the identical initial generated candidate across conditions (shared_initial_candidate semantics) we isolate only downstream guarded-stage effects (emulation observations or permitted repair invocations) rather than conflating them with generation variability.

Gaps and evaluation needs. While surveys and system reports argue for generate→validate→repair and for emulator-assisted validation as best practice for high-assurance NetOps, they do not provide large-scale, preregistered paired estimates of the marginal detection benefit under controlled generation semantics. The literature also documents important open questions that our study was designed to probe: how much additional coverage does sandboxed emulation provide when a strong deterministic validator already targets the same workflow patterns, and which task regimes (adversarial inputs, cross-vendor CLI heterogeneity, or stateful multi-step procedures) are most likely to reveal nontrivial guarded-stage benefits? By situating our paired confirmatory design amid these prior findings, we provide an artifact-grounded quantitative datapoint addressing one narrow slice of the broader evaluation agenda advocated in [1]–[3].

III. METHODOLOGY

Preregistration and analytic intent. We preregistered a confirmatory paired-binary design (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e). The primary estimand was the paired_success_rate_difference_guarded_minus_baseline aggregated across confirmatory samples; the preregistered minimum effect-of-interest for power planning was 0.20 absolute. The preregistration (frozen prior to execution) specified: inclusion/exclusion rules, deterministic missingness and reseed policies (up to two automatic reseeds for parser failures), repair-stage limits (≤ 1 repair call per task), contamination detection and deterministic replacement rules, and the analysis executor (paired_binary_analysis_v1) with paired inference procedures and pre-registered sensitivity analyses.

Task bank, topology templates, and vendor abstraction. Confirmatory items were synthetic intent→configuration tasks generated from the intent_configuration_repair_v1 family. The bank covered three abstracted vendor-CLI families and three topology templates (leaf–spine, 3-node service chain, simple triangle). Each task manifest encoded: the initial device state, an explicit intent, a required command sequence, allowed-touched fields, and parser-acceptance constraints. Inclusion required vendor-specific parser acceptance after up to two deterministic reseeds; irrecoverable parser failures were to be deterministically excluded and replaced per the preregistered missingness plan. In the executed run there were zero irrecoverable parser exclusions and complete_pair_count = 720 (analysis/missingness_summary.csv).

Model interface, sampling semantics, and shared_initial_candidate enforcement. The declared/requested model was recorded as "gpt-5-mini" and the execution model_configuration.json contains declared_model_version = "gpt-5-mini", max_output_tokens = 2000, and temperature = null (sha256 a40e96e212ab87da...). Important operational clarification (preregistered and enforced): temperature = null in the archived configuration indicates the client did not set an explicit temperature parameter; null does not imply temperature=0 or provider-side determinism. To isolate guarded-stage effects we enforced shared_initial_candidate

semantics: for each pair the pipeline performed a single sampled initial generation (shared) which both the baseline validator and the guarded pipeline consumed. Concretely, provider_call_audit and deterministic_reconciliation artifacts record stage_counts: shared_initial_generation = 720 and repair = 1, and model_calls_used = 721 (720 shared + one repair-stage call). Because the same sampled candidate was analyzed by both conditions for each pair, any between-condition discordance must stem from (i) guarded-stage sandboxed emulation observations not encoded as validator violations, and/or (ii) the permitted repair-stage intervention recorded in run accounting.

Validator implementation and scoring rule. Baseline validation used deterministic_netops_validator_v5 (scorer_version = 5.0) executing the preregistered scoring rule full_intent_constraint_validation. The validator performs multi-level checks recorded in per-episode validator_trace objects: vendor-specific parsing, syntactic command parsing, command ordering/workflow checks against required_sequence, required-field presence, dependency constraints (e.g., must-shutdown-before-mtu-change patterns), and policy constraints (protected interfaces, group-wise minimum-active constraints). The validator emits structured outputs used in scoring: normalized_configuration, parsed_command_count, required_command_count, observed_touched_field_count, violation_count, and violation_codes. Per-episode artifacts (execution/scoring/task-XXXXX-*.json) show these fields and validator_version = 5.0 for all confirmatory episodes.

Sandboxed-emulation harness and guarded pipeline semantics. The guarded pipeline applied the baseline deterministic validator and then, for candidates not blocked by parser failure, executed an automated sandboxed-emulation harness that instantiated an emulated representation of the declared topology and applied the candidate commands. The emulation harness produced runtime observations that were normalized and summarized into the same validator-trace structure (validation_before/validation_after, final_state, parsed_command_count, observed_touched_field_count). The emulation stage was designed to detect runtime-only symptoms such as transient ordering-dependent state changes, reachability/regression signals, or state transitions that static rules do not explicitly capture. The preregistered repair policy permitted up to one repair-stage model call per task; repair calls (if used) and their provider traces are recorded in the execution manifest and deterministic_reconciliation artifacts.

Inclusion/exclusion, missingness, and contamination controls. The preregistered missingness plan allowed two automatic deterministic reseeds for parser failures; irrecoverable parser failures would be deterministically replaced. Contamination controls included deterministic SHA-256 artifact hashing, automated prompt-template divergence checks, and validator/emulator binary/config checksum verification before confirmatory execution. The execution manifest and analysis/contamination_summary.csv report no contamination flags and no irrecoverable parser exclusions for the confirmatory

set.

Statistical analysis procedures. The primary confirmatory analysis used the `paired_binary_analysis_v1` executor specified in the preregistration. The primary test computed the aggregate paired detection-rate difference (guarded - baseline) across complete pairs and reported an exact McNemar two-sided p-value. Pre-registered uncertainty quantification used paired-bootstrap percentile confidence intervals with 10,000 resamples and a fixed seed (`bootstrap_seed = 1729`) for reproducibility. Secondary exploratory analyses (per-class and per-vendor paired differences) were pre-registered with Bonferroni adjustments; these are archived but not the focus of the confirmatory report. The preregistered default treatment of irrecoverable technical failures for the primary analysis was to exclude complete-pair technical losses (`complete_pair_primary`); a sensitivity analysis treats irrecoverable failures as nondetections (documented in the preregistration). The executed analysis followed the preregistered `complete_pair_primary` plan; analysis artifacts and logs are archived (`analysis/analysis_log.jsonl` and `analysis/paired_contingency_table.csv`).

Deterministic reconciliation and run accounting. To enable artifact-grounded claims and independent checking we recorded `provider_call_audit` and `deterministic_reconciliation` artifacts. These reconcile: `planned_episode_count = 1440`, `completed_episode_count = 1440`, `complete_pair_count = 720`, `failed_episode_count = 0`, `model_calls_used = 721`, and `stage_counts` (`shared_initial_generation = 720`; `repair = 1`). These machine-verifiable run-accounting artifacts are referenced throughout the manuscript when we state numerical outcomes and when tracing the single discordant guarded-only detection back to a documented repair-stage invocation.

IV. RESULTS

Primary confirmatory counts and test statistics (artifact-grounded). The preregistered paired analysis produced the following exact contingency counts (`analysis/paired_contingency_table.csv`, sha256 efe6e8e7016fa843a8226e911dbde829e943903d719101da8d956b3ff899133c): `n11 = 719` (detected by both), `n10 = 1` (guarded-only), `n01 = 0` (baseline-only), `n00 = 0` (neither). Marginal detection rates are: `baseline = 719/720 = 0.9986111111111111` and `guarded = 720/720 = 1.0`. The paired estimate (guarded - baseline) = `0.00138888888888888884`. The exact McNemar two-sided test reported `p = 1.0`. Paired-bootstrap percentile 95% CI (10000 resamples, `bootstrap_seed = 1729`) = `[0.0, 0.004166666666666667]`. All numeric values above are reproduced verbatim from `analysis/results.json` and analysis artifacts.

Bootstrap and test details. The bootstrap CI was computed with 10000 resamples (`seed = 1729`) using paired resampling of episode-pairs; the analysis executor used a paired-bootstrap-percentile approach (artifact: `analysis/analysis_log.jsonl`). The McNemar exact two-sided p-value complements the bootstrap CI by testing symmetry in the discordant-pair counts; here the observed discordant counts (`n10 = 1`, `n01 = 0`) produce no evidence of a systematic guarded advantage at conventional

significance thresholds (`p = 1.0` under the exact test reported by the executor).

Run accounting and discordant-pair provenance. Provider-call accounting and deterministic reconciliation confirm `model_calls_used = 721` (stages: `shared_initial_generation = 720`; `repair = 1`) and `complete_pair_count = 720` (`analysis/deterministic_reconciliation.json`, sha256 49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa). Because `shared_initial_candidate` semantics were enforced, the single guarded-only detection (the unique `n10` entry) must arise from downstream guarded-stage activity (sandboxed emulation) or from the single recorded repair-stage invocation.

Explicit archival pointer to the discordant episode (artifact-grounded). The unique discordant episode can be located in the archived raw-results file: `execution/raw_results.jsonl` (input-results SHA-256 = `df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8`). Within that file, filter for the entry where `baseline_score = 0` and `guarded_score = 1`; that `raw_results.jsonl` record contains the episode's `pair_id` and the linked `validator_trace` and any `repair_provider_trace` fields that document the repair-stage invocation. The deterministic reconciliation artifact (`analysis/deterministic_reconciliation.json`, sha256 49731a2e52e0...) and the `provider_call_audit` block therein corroborate that the repair-stage call recorded in provider accounting maps to that unique discordant pair. We provide these exact artifact identifiers so readers can unambiguously locate the discordant pair and its repair/emulation traces in the archived bundle (see Reproducibility pointers below).

Discordant attribution and repair linkage. The archived `raw_results` entry for the discordant pair includes `validator_trace` (`validation_before/after`), any emulation observations, and fields capturing whether a repair prompt was issued and its trace path. The `deterministic_reconciliation.json` `provider_call_audit` documents `stage_counts` and confirms exactly one repair-stage call; matching timestamps and trace-path fields in `raw_results.jsonl` link that repair-stage call to the guarded-only detection. Thus the run artifacts substantiate that the single guarded-only detection corresponds to downstream guarded-stage activity (repair/emulation) rather than to generation differences, consistent with `shared_initial_candidate` semantics.

Representative per-episode diagnostics. Representative scoring traces archived under `execution/scoring/task-000001-*.json ... task-000006-*.json` include detailed `validator_trace` objects with `normalized_configuration`, `validation_before/after` states, `parsed_command_count`, `required_command_count`, `violation_count`, and `validator_version = 5.0`. Those representative artifacts illustrate that the vast majority of `shared_initial` candidates complied with validator rules (`violation_count = 0`) and that sandboxed emulation generally corroborated validator findings in this curated task bank. The `analysis/condition_summary.csv` artifact (sha256 394bc690671b7...) summarizes successes/failures by condition and reproduces the marginal success rates reported above.

Synthesis of numeric outcomes. In short, the guarded pipeline detected one more successful item than the baseline (720 vs. 719) across 720 paired episodes. The paired estimate (~ 0.00139) and its 95% bootstrap CI ($[0.0, 0.0041667]$) are both orders of magnitude smaller than our preregistered minimum effect-of-interest (0.20). Provider-call accounting (`model_calls_used = 721`, `repair = 1`) further shows that the observed discordance is attributable to recorded downstream activity rather than to between-condition generation variance. These concrete, artifact-grounded outcomes are the basis for the interpretations and operational boundary conditions we discuss elsewhere in the manuscript.

V. DISCUSSION

Why was the guarded-stage aggregate benefit negligible? The archived evidence supports three artifact-grounded factors.

1) Validator ceiling effect. `Deterministic_netops_validator_v5` covered the syntactic and explicit semantic properties encoded in the synthetic task bank (shutdown—configure—restore, make-before-break uplink switchover, paired MTU changes). Baseline success rate ≈ 0.9986 indicates near-universal validator coverage for these curated patterns, leaving negligible headroom for the guarded stage.

2) Task curation and inclusion policy. Confirmatory items were parser-validated synthetic tasks specifically designed to exercise workflow policies but not constructed adversarially to evade validator checks. This curation concentrates cases inside the validator’s coverage envelope and reduces the prevalence of subtle runtime-only failure modes that emulation might reveal.

3) `Shared_initial_candidate` semantics and limited repair activity. Enforcing one sampled initial candidate per pair isolates guarded-stage contributions but removes between-condition generation variability. `Provider_call_audit` and `deterministic_reconciliation.json` record one repair-stage invocation aligned with the single `n10` discordant count; thus the observed discordance is attributable to downstream guarded-stage activity recorded in run accounting rather than generation differences.

Operational implications and boundary conditions. For environments where a high-coverage static validator exists and inputs match its expected patterns (parser-validated, constrained intents), adding automated sandboxed emulation may provide negligible aggregate additional detection at scale. This conditional conclusion must not be generalized: in operational regimes with noisier natural-language intents, adversarial inputs, broader vendor diversity, stateful multi-step interactions, or validator gaps, guarded runtime checks and higher-fidelity emulation are likely to provide substantive benefit. The archived per-episode traces support targeted follow-up hypotheses: adversarial task injection, broader vendor-CLI coverage, and complementary validator families should increase the guarded-stage marginal contribution.

VI. LIMITATIONS

- Synthetic task bank and deterministic task generator produce limited ecological diversity and create a validator-ceiling effect; results may not generalize to noisier, adversarial, or production distributions.
- Single declared/requested model family (`gpt-5-mini`) and single validator implementation (`deterministic_netops_validator_v5`) limit generalizability across models and validator families.
- Preregistered `shared_initial_candidate` semantics (one initial generation reused across paired conditions) isolate guarded-stage contribution but remove between-condition generation variability; this design choice constrains discordance sources to downstream guarded-stage observations or repair calls.
- Sandboxed-emulation fidelity relative to vendor/OS production behaviors and large-scale stateful interactions is limited by the emulation harness used; higher-fidelity emulators could change outcomes in other regimes.
- Study power planning targeted a minimum detectable paired difference of 0.20; the observed aggregate effect (~ 0.0014) lies far below that design sensitivity and should be interpreted as a narrowly scoped near-zero finding for this experimental regime rather than a general negative result for all NetOps workflows.
- Provider-side nondeterminism and hidden caching remain residual uncertainties despite deterministic reconciliation procedures; these are documented in `analysis/deterministic_reconciliation.json` and `provider_call_audit` artifacts.

VII. CONCLUSION

We preregistered and executed a paired confirmatory study comparing a strong deterministic validator with the same validator augmented by sandboxed emulation on a curated synthetic intent→configuration task bank. Over 720 paired tasks the guarded pipeline produced a negligible aggregate detection gain (0.0013888888888888884) with exact McNemar $p = 1.0$ and 95% bootstrap CI $[0.0, 0.004166666666666667]$. Run accounting shows one repair-stage invocation corresponding to the single guarded-only detection; because `shared_initial_candidate` semantics were enforced, archived per-episode traces permit independent verification of that mapping via `execution/raw_results.jsonl` and `analysis/deterministic_reconciliation.json`. We interpret the result as arising from validator ceiling effects and curated task design; follow-up work should focus on adversarial and production-like task banks, multi-validator ensembles, and higher-fidelity emulation to identify regimes where guarded stages add substantive operational value.

Reproducibility pointers (compact). The canonical execution bundle archived by the run contains: `execution/execution_manifest.json` (execution summary and preregistration SHA-256), `execution/raw_results.jsonl` (input-results SHA-256 `df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8`),

analysis/deterministic_reconciliation.json (sha256 49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa), execution/model_configuration.json (sha256 a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd820), analysis/paired_contingency_table.csv (sha256 efe6e8e7016fa843a8226e911dbde829e943903d719101da8d956b3ff899133c), and analysis/condition_summary.csv (sha256 394bc690671b731194f9b42b4dcbe28fdd6e2ec8cf898b79831225343c22c28).

The discordant episode is the unique raw_results.jsonl entry with baseline_score = 0 and guarded_score = 1; that record contains the pair_id and validator/emulation traces that map it to the single repair-stage invocation recorded in deterministic_reconciliation.json. These artifacts are archived as the canonical reproducibility bundle referenced by the execution manifest and the preregistration record. (See Disclosure for exact manifest references.)

DISCLOSURE STATEMENT

Disclosure: Autonomous post-lock research run executed under the frozen master prompt supplied to the system. Declared/requested model: "gpt-5-mini" (provider recorded as openai_responses). Deterministic validator: deterministic_netops_validator_v5. Orchestration adapter family: hosted_netops_gvr_v1. Preregistration id: f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e (preregistration SHA-256: 0169919fb8c5aedb2c5a78e031f1ab5c3aeda405cde1ae80fb2f864768595c1b; recorded in the execution manifest). Initial master-prompt immutable reference: provenance/master_prompt.txt, SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba. Execution summary (artifact-grounded): completed_episode_count = 1440, complete_pair_count = 720, model_calls_used = 721 (720 shared initial + 1 repair-stage call); stage_counts and provider-call audit are archived in analysis/deterministic_reconciliation.json (sha256 49731a2e52e0...). Human role and autonomy boundary: a human Research Director selected the broad NetOps topic family and constrained the pre-lock master prompt; after the autonomy lock the agentic pipeline autonomously performed literature selection, preregistration, code and prompt generation, experiment execution, analysis, peer review, and manuscript drafting. No manual human editing of technical manuscript content occurred after the autonomy lock. The execution manifest and archived artifacts named above serve as the canonical reproducibility bundle.

REFERENCES

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